

LATEST CHIPS Changing The AUTOMOBILE INDUSTRY

DEEPSHIKHA SHUKLA

This article discusses the latest chips that are transforming the automobile industry

For many years, the internal combustion engine has been at the heart of the automobile. In the last few years, tiny slices of silicon have increasingly been acting as its brain, telling the engine what to do and when. In this new era, microprocessors and software that run automated and electric vehicles (EVs) have become ever more important.

Power device technologies such as silicon-carbide metal oxide semiconductor field effect transistors (SiC MOSFETs), diodes and high-voltage insulated-gate bipolar transistors (IGBTs) improve the performance of EV systems and charging infrastructure. Multi-tasking, application-specific LSI (system ASICs) controllers reduce complexity and size, and improve efficiency of automotive systems.

Automotive green solutions include LED lighting that reduce energy consumption and extend the life of the lamp (generating less waste). SiC products (MOSFETs and Schottky barrier diodes) improve efficiency and robustness of the systems.

The aim now is to develop components for simplifying system designs, and improving efficiency and reliability. Safety and

reliability are paramount for automotive systems.

The focus is also on designing development systems that reduce emissions and help original equipment manufacturers (OEMs) meet demanding standards like BS-VI, for both two and four wheelers.

Focus is also on LSI (ASIC) devices for making control systems simpler and more compact.

Steps to ensure quality and reliability of automotive components are:

1. At chip level, develop chip design rules like clearances and routing to improve reliability.
2. Set up separate wafer processing facilities for automotive products.
3. Modify packaging designs to reduce stresses on wire bonds, leads and solder joints, to improve system-level reliability.
4. Stick to relevant AEC standards (Q100, Q101) for all automotive parts.

Latest chips for automotives

Satish R.M, automotive product strategy manager, ROHM Semiconductors, says, "ROHM obtained ISO 26262 process certification, which is an international standard for functional safety for automotive products. Focus applications include electronic fuel injection systems (fuel pump controls, etc), regulators/rectifiers, integrated starter generators, body electronics (lighting, displays, etc), traction motor inverters, DC-to-DC controllers, onboard chargers and charging stations for EVs, and body control modules (lighting, mirror control, power windows, etc), among others."

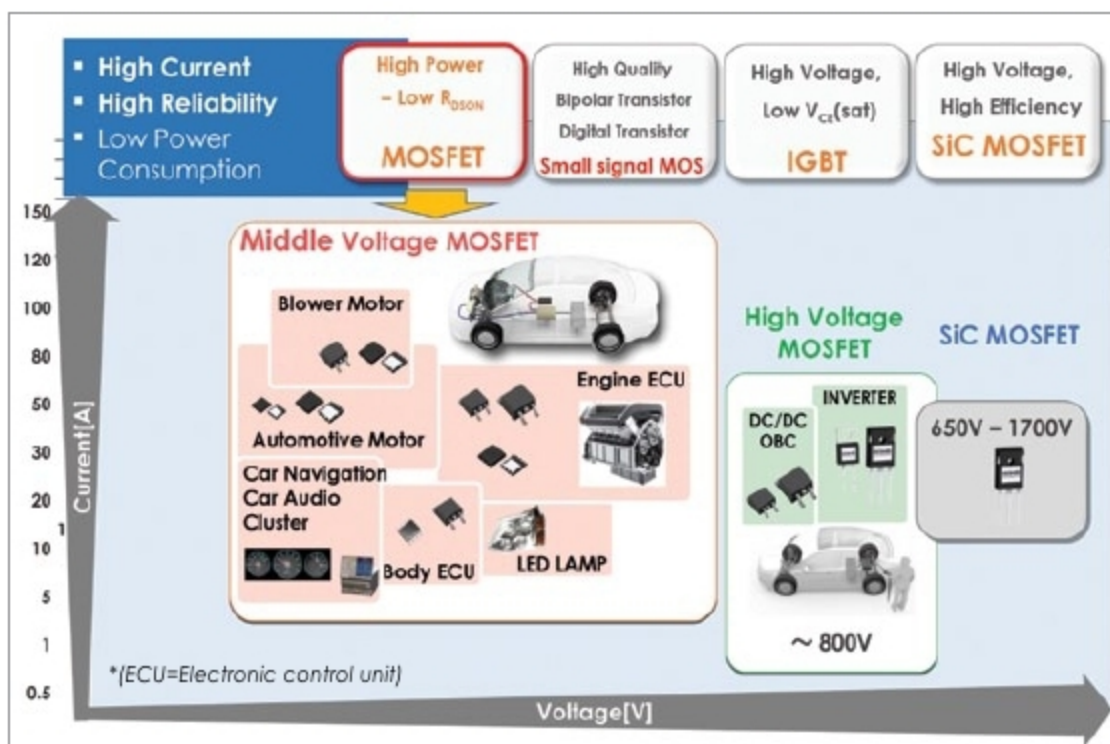


Fig. 1: Device technology for automotive applications (Credit: Satish R.M., ROHM Semiconductor)

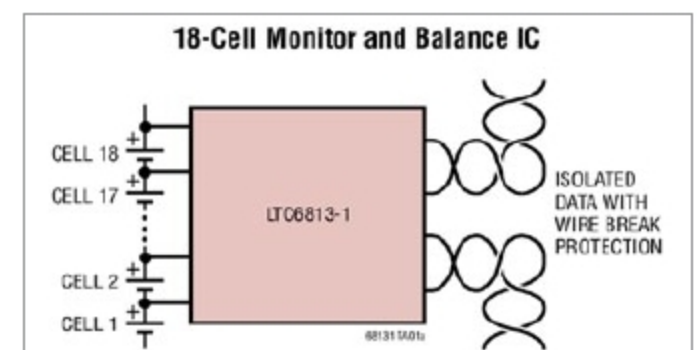


Fig. 2: IC LTC6813-1 (Credit: www.analog.com)